A1-SP-F1 ($\tilde{\lambda}$ = 0.1) on the Ψ_A1 metric; contours = distress score (LightGBM, out-of-fold, 30-NN mean in 2-D)

(a) Class 2: Classical MDS of Ψ_A1 (2-D share 30%)

(b) Class 1: Classical MDS of Ψ_A1 (2-D share 30%)

(c) Class 2: Distress logit (LogReg) × MDS 1 residual (Ψ_A1)

(d) Class 1: Distress logit (LogReg) × MDS 1 residual (Ψ_A1)

